TCAD

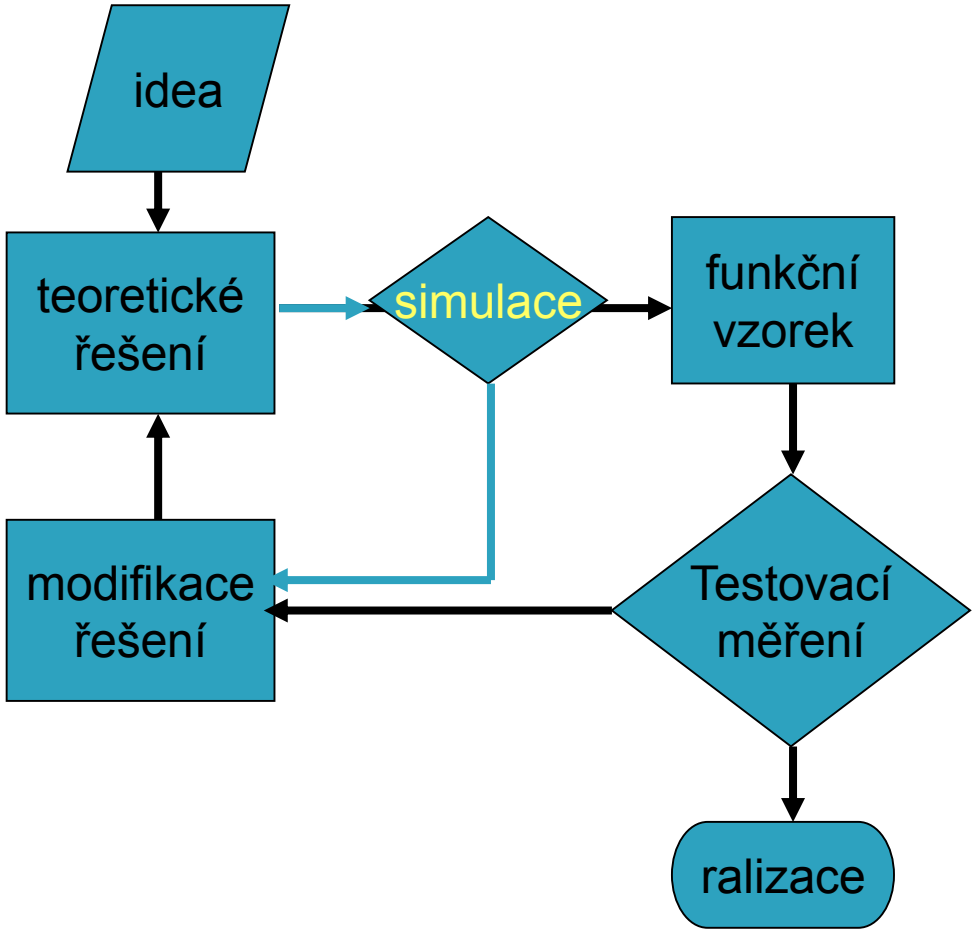
TECHNOLOGY COMPUTER AIDED DESIGN

- Vyhodnocení modifikovaných technologií
- Citlivostní analýzy technologických parametrů
- Porozumění fyzikálním jevům
- Predikce elektrických parametrů
- Spolehlivostní analýza součástek a struktur
- Extrakce a optimalizace parametrů pro obvodový návrh
- Extrakce parazitních parametrů pro signálovou analýzu
- Charakterizace technologických knihoven

Návrhový cyklus

simulace

podstatné zkrácení
návrhového cyklu
+
úspora materiálu

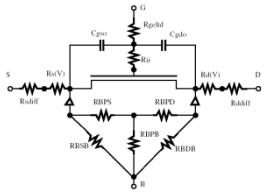


Simulační úrovně

proudy a napětí v obvodu

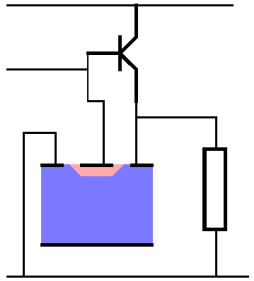
elektrické schéma

SIMULACE ELEKTRONICKÝCH OBVODU



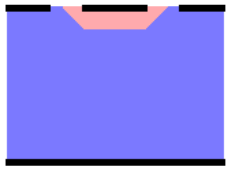
SMÍŠENÁ SIMULACE

elektrické parametry součástek



napětí, proudy na elektrodách

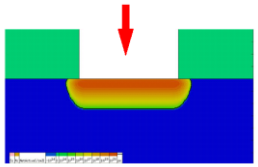
SIMULACE ELEKTRONICKÝCH SOUČÁSTEK



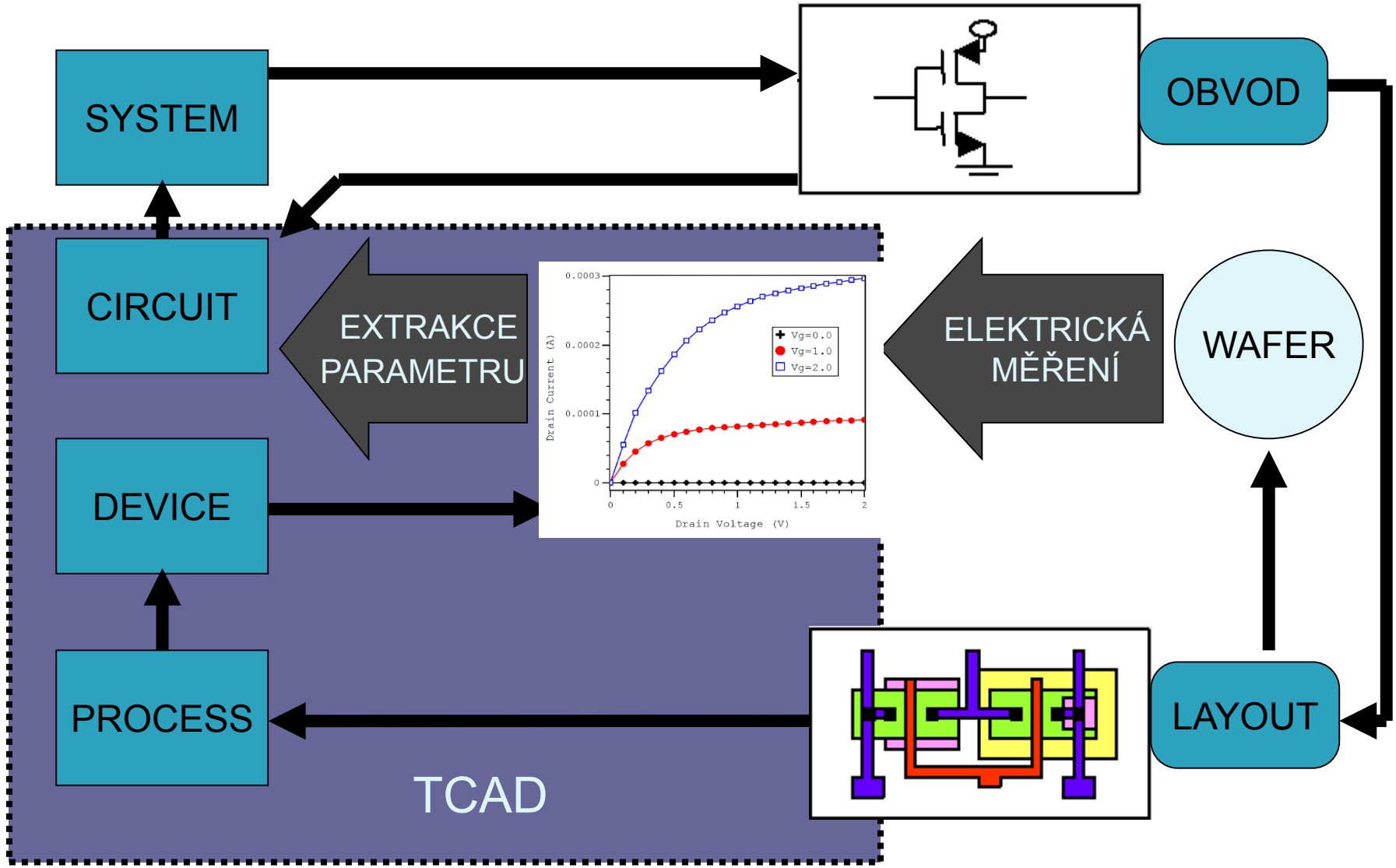
geometrie, koncentrační profily

technologické parametry

TECHNOLOGICKÁ SIMULACE



Simulace ve výrobním procesu



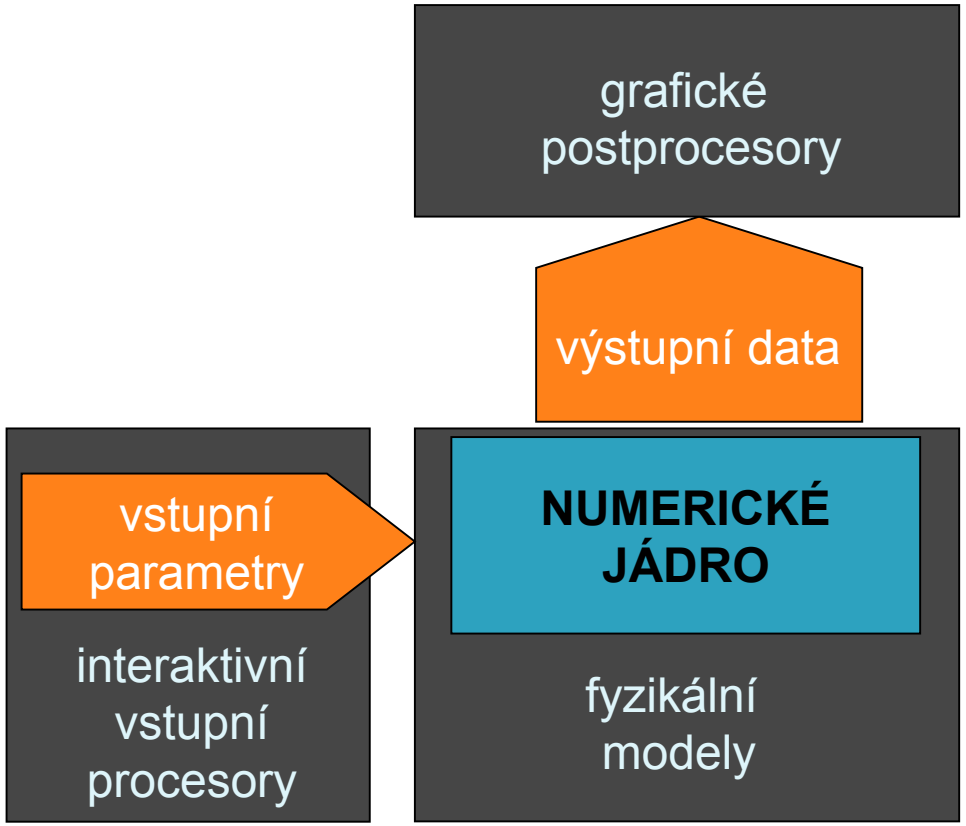
Simulační program

Použité fyzikální modely aproximují reálné děje ve struktuře, součástce nebo obvodu

požadavky na přesnost
rozměry struktur
rychlost dějů

↓

úroveň fyzikálního popisu



Řešené rovnice

1 – 6 parciálních diferenciálních rovnic

Keyword	Corresponding equation	Corresponding variable
carriers = 0	Poisson	Electrostatic potential
carriers=1 electron	Electron continuity	Electron density
carriers=1 hole	Hole continuity	Hole density
lat.temp	Temperature	Temperature
htce	Electron temperature	Electron temperature
htch	Hole temperature	Hole temperature

Diskretizace

- analytické řešení parciálních diferenciálních rovnic

  jen ve speciálních případech

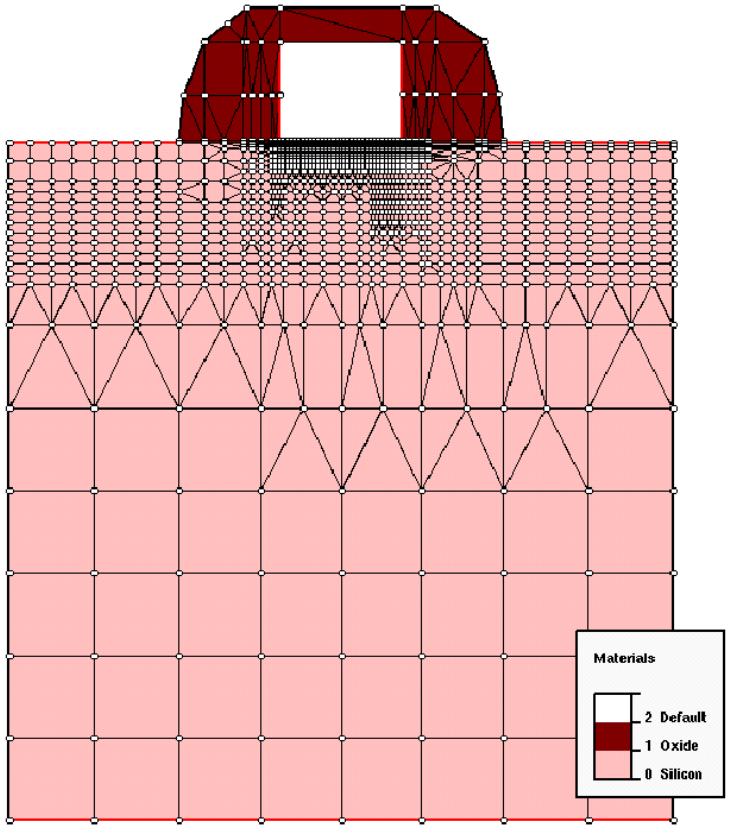
- numerické řešení

  aproximativní –

  nahrazení derivací diferencemi

  rozdělení řešené oblasti pomocí sítě

  jemnější síť → přesnější řešení



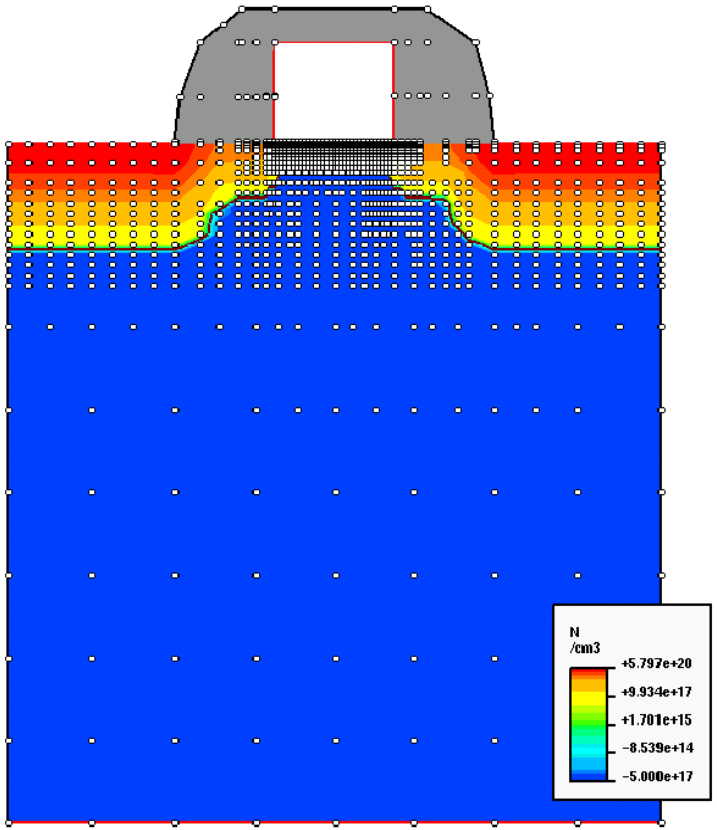
Diskretizace

- hledání řešení pouze v diskrétních bodech – uzlech :

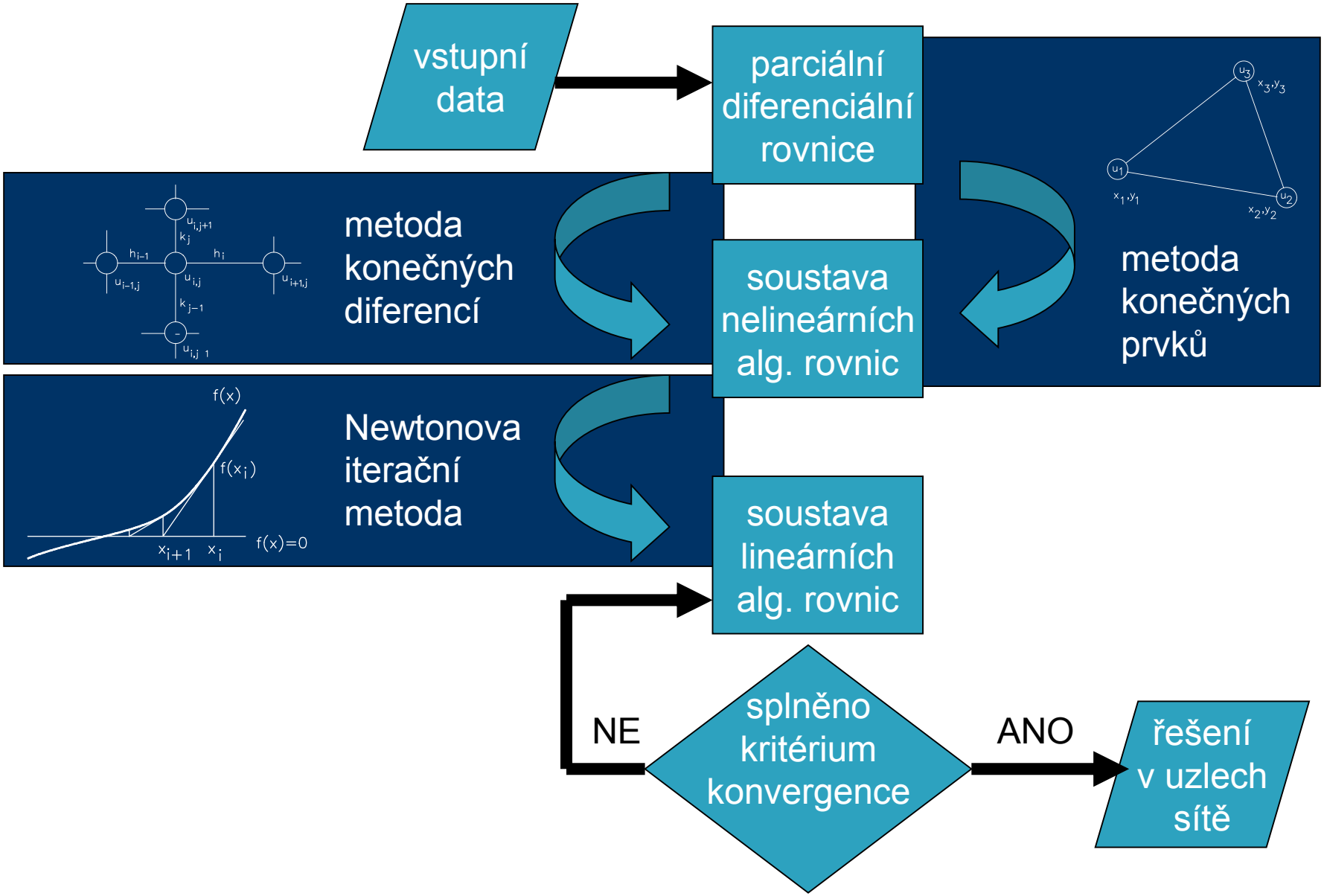
  $\longrightarrow \quad u_{i,j}$

- řešení ve všech bodech pomocí interpolace

  $\longrightarrow \quad u(x,y)$



Numerické řešení diferenciálních rovnic



Metoda konečných diferencí

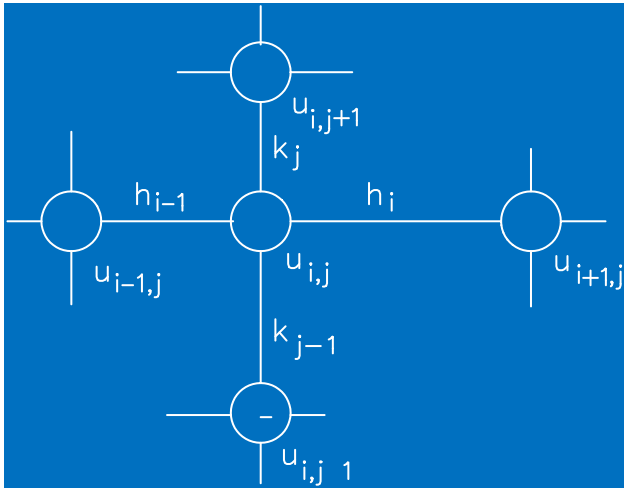
- 2D síť pro metodu konečných diferencí:

  $u(x,y) \longrightarrow u_{i,j}$
  $i = 1..M,\ j = 1..N$
  M x N uzlů

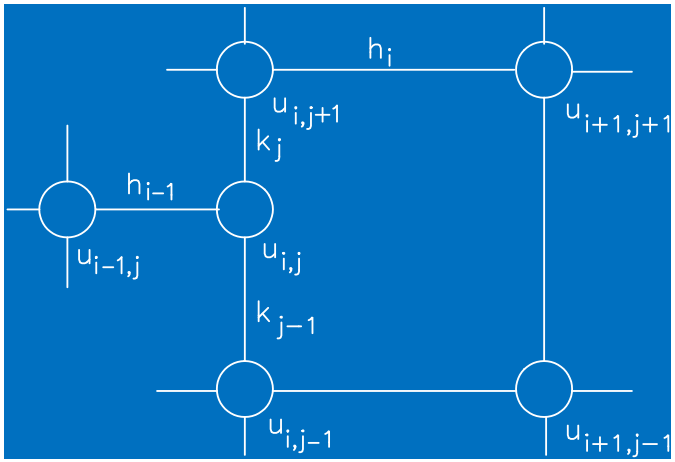
  2D síť pro metodu konečných objemů:

Metoda konečných diferencí

- okolí uzlu u(i,j) v síti pro metodu konečných diferencí



- okolí uzlu u(i,j) v síti pro metodu konečných objemů



Metoda konečných diferencí

- Taylorův rozvoj hledané fukce v okolí uzlu:

$$u_{i+1,j} = u_{i,j} + h_i \left(\frac{\partial u}{\partial x} \right)_{i,j} + \frac{h_i^2}{2!} \left(\frac{\partial^2 u}{\partial x^2} \right)_{i,j} + \frac{h_i^3}{3!} \left(\frac{\partial^3 u}{\partial x^3} \right)_{i,j} + \dots$$

$$u_{i-1,j} = u_{i,j} - h_{i-1} \left(\frac{\partial u}{\partial x} \right)_{i,j} + \frac{h_{i-1}^2}{2!} \left(\frac{\partial^2 u}{\partial x^2} \right)_{i,j} - \frac{h_{i-1}^3}{3!} \left(\frac{\partial^3 u}{\partial x^3} \right)_{i,j} + \dots$$

Metoda konečných diferencí

- náhrada derivací diferencemi:

$$\left(\frac{\partial u}{\partial x}\right)_{i,j} = \frac{u_{i+1,j} - u_{i-1,j}}{h_i + h_{i-1}}$$

$$\left(\frac{\partial^2 u}{\partial x^2}\right)_{i,j} = \frac{u_{i+1,j}\, h_{i-1} + u_{i-1,j}\, h_i - u_{i,j}(h_{i-1} + h_i)}{(h_i + h_{i-1})\, h_{i-1}\, h_i}$$

→ M x N nelineárních algebraických rovnic (z 1 diferenciální rovnice)

Metoda konečných prvků

aproximace bilineárním polynomem:

$$u(x,y) = a_0 + a_1 x + a_2 y$$

transformace souřadnic:

$$x = x_1 + (x_2 - x_1)\xi + (x_3 - x_1)\eta$$

vyjádření hledané funkce pomocí hodnot ve vrcholech:

$$u(\xi,\eta) = u_1 + (u_2 - u_1)\xi + (u_3 - u_1)\eta$$

Metoda konečných prvků

aproximace řešení v prvku pomocí tvarových funkcí:

$$\phi_1 = 1 - \xi - \eta$$
$$\phi_2 = \xi$$
$$\phi_3 = \eta$$

$$u(\xi,\eta) = \sum_{i=1}^{3} u_i \phi_i(\xi,\eta)$$

aproximace v celé oblasti:

$$u = \sum_{k=1}^{n} \left(\sum_{i=1}^{3} u_i \phi_i \right)_k$$

Metoda konečných prvků

- slabá formulace diferenciální rovnice:

$$\int_D V \,.\, F(u)\, dx + \int_{\partial D} v \,.\, f(u)\, dA = 0$$

libovolné ortogonální funkce

$V = V_\nu\,,\ v = v_\nu\,,\ \nu = 1, N$

- Galerkinova metoda – v .. tvarové funkce



řešení soustavy nelineárních algebraických rovnic

Hlavní dodavatelé TCAD systémů

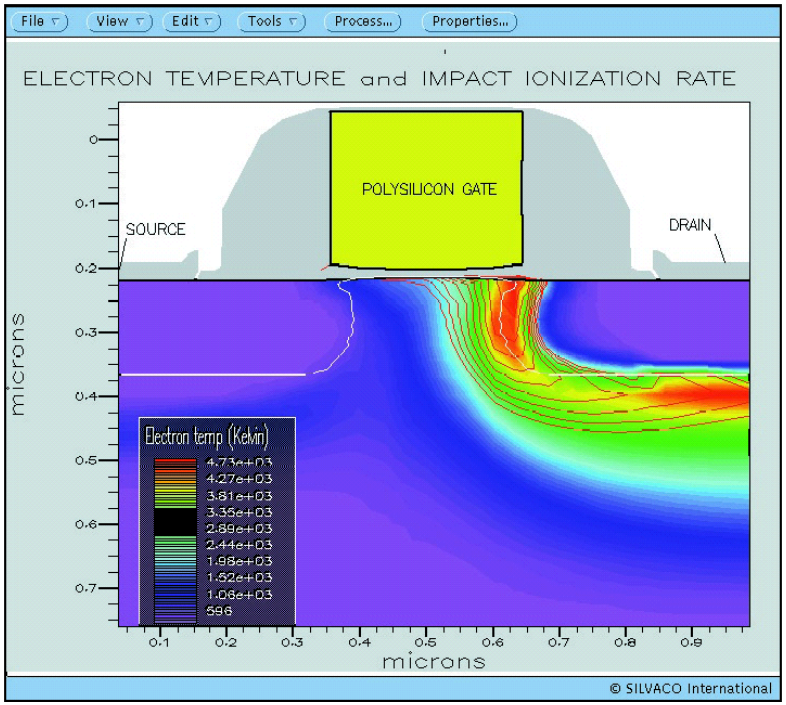
- Integrated Systems Engineering (ISE)



- Synopsys (Technology CAD)
- Silvaco International
- PDF Solutions
- SEQUOIUA Design Systems
- Siborg Systems, Inc.

SILVACO International

VWF – virtuální polovodičová továrna



ATHENA 1D/2D Process Simulation Software

SSuprem3 1D Process Simulation Software

ATLAS Device Simulation Software

MERCURY FastFET Simulation Software

Mocasim Monte Carlo Transport Parameter Generator

CLEVER Physics-based Parasitic Extraction Software

UTMOST SPICE Modeling Software

Virtual Wafer Fab Interactive Environment

DeckBuild Run-Time Environment

DevEdit Structure & Mesh Editor

TonyPlot 1D/2D Interactive Visualization Utility

TonyPlot 3D Interactive Visualization Utility

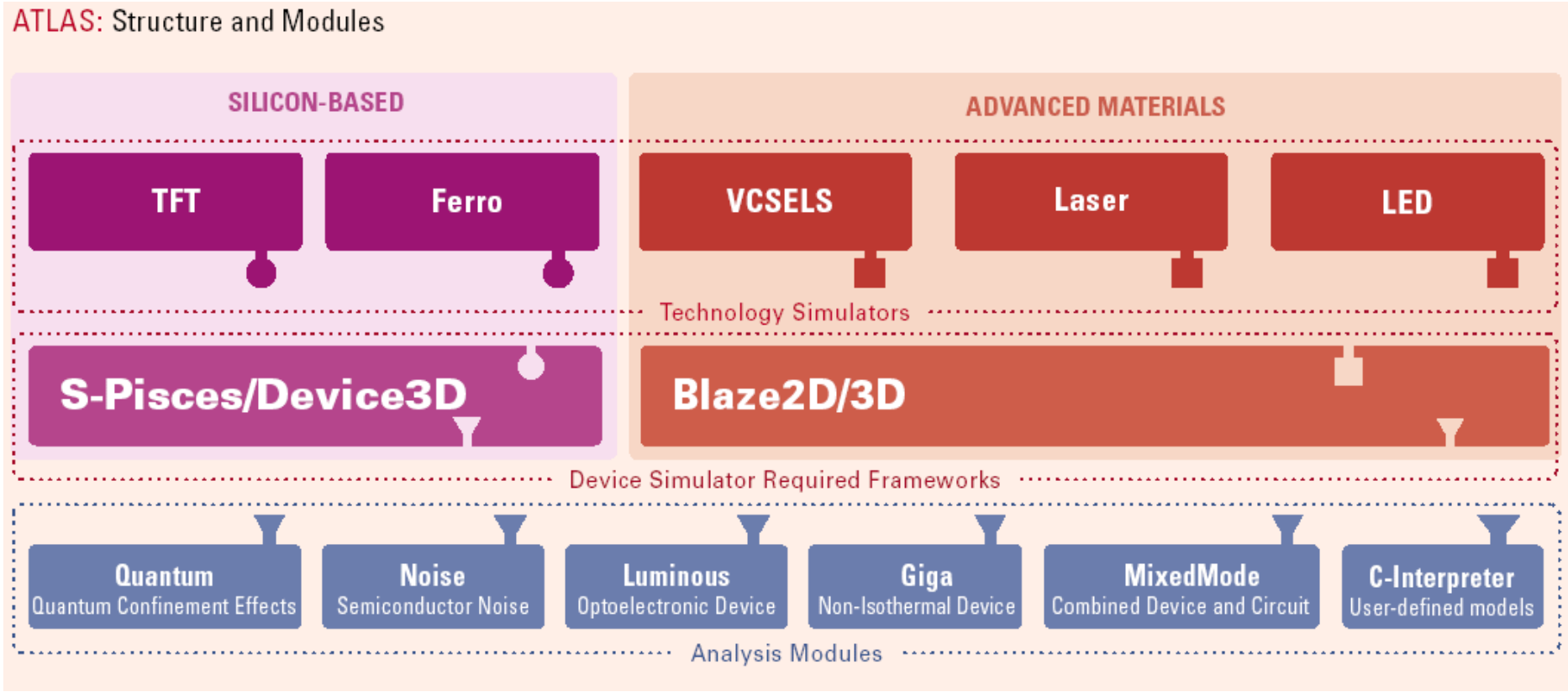
C-Interpreter User Defined Models

Optimizer Utility

MaskViews Layout Editor

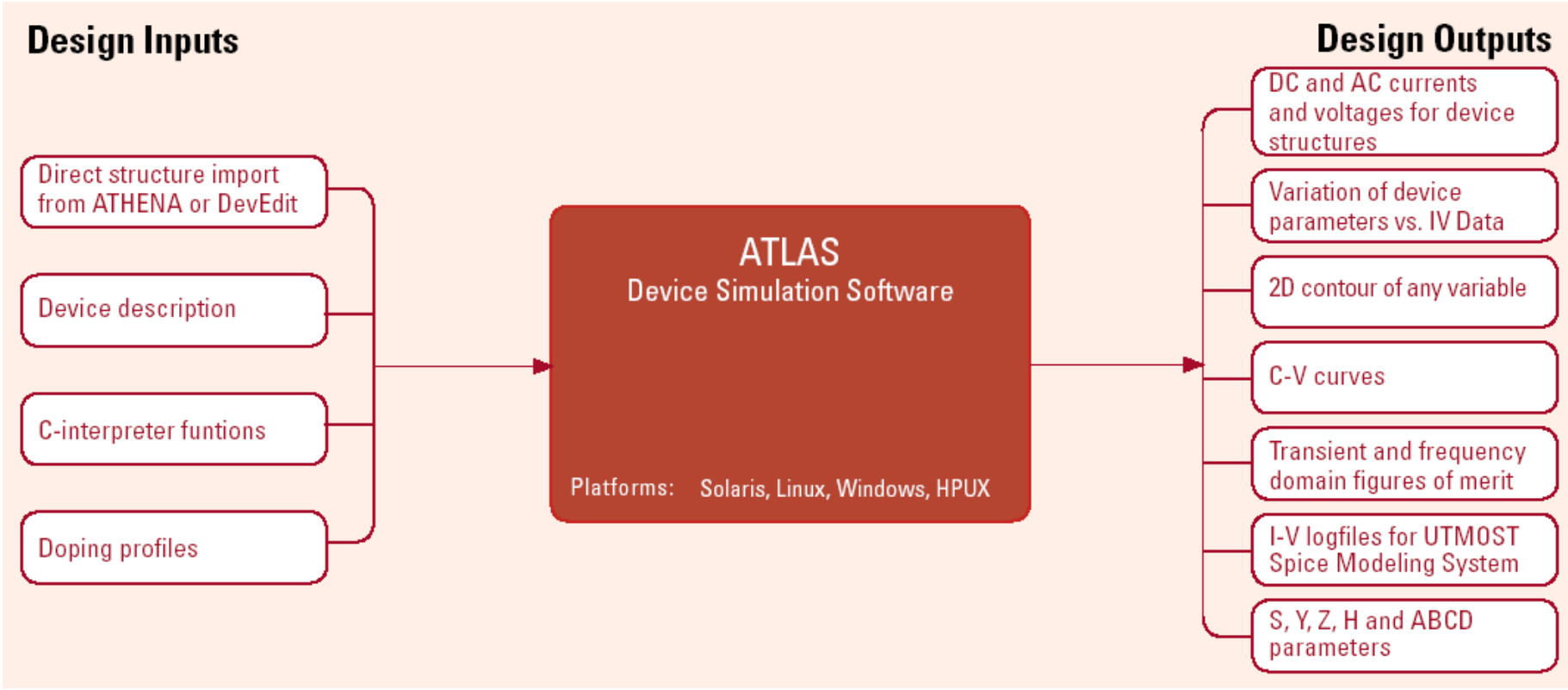
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Device simulation – ATLAS – struktura a moduly



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Device simulation – ATLAS – vstupy / výstupy



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Table 2-1. Carrier Statistics Models		
Model	**Syntax**	**Notes**
Boltzmann	BOLTZMANN	Default model
Fermi-Dirac	FERMI	Reduced carrier concentrations in heavily doped regions (statistical approach)
Incomplete Ionization	INCOMPLETE	Accounts for dopant freeze-out. Typically used at low temperatures.
Silicon Ionization Model	IONIZ	Accounts for full ionization for heavily doped Si. Use with INCOMPLETE.
Bandgap Narrowing	BGN	Inportant in heavily doped regions. Critical for bipolar gain. Use Klaassen Model.

Model	Syntax	Notes
Concentration Dependent	CONMOB	Lookup table valid at 300K for Si and GaAs only. Uses simple power law temperature dependence.
Concentration and Temperature Dependent	ANALYTIC	Caughey-Thomas formula. Tuned for 77-450K.
Arora's Model	ARORA	Alternative to ANALYTIC for Si
Carrier-Carrier Scattering	CCSMOB	Dorkel-Leturq Model. Includes n, N and T dependence.. Important when carrier concentration is high (e.g., forward bias power devices).
Parallel Electric Field Dependence	FLDMOB	Si and GaAs models. Required to model any type of velocity satuation effect.
Tasch Model	TASCH	Includes transverse field dependence. Only for planar devices. Needs very fine grid.
Watt Model	WATT	Transverse field model applied to surface nodes only.
Klaassen Model	KLA	Includes N, T and n dependence. Applies separate mobility to majority and minority carriers. Recommended for bipolar devices
Shirahata Model	SHI	Includes N, $E_{\perp}$. An alternative surface mobility model that can be combined with `KLA`.
Modified Watt	MOD.WATT	Extension of `WATT` model to non-surface nodes. Applies constant $E_{\perp}$ effects. Best model for planar MOS devices
Lombardi (CVT) Model	CVT	Complete model including N, T, E// and $E_{\perp}$ effects. Good for non-planar devices.
Yamaguchi Model	YAMAGUCHI	Includes N, E// and $E_{\perp}$ effects. Only for 300K.

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Table 2-3. Recombination Models		
Model	**Syntax**	**Notes**
Shockley-Read-Hall	SRH	Uses fixed minority carrier lifetimes. Should be used in most simulations.
Concentration Dependent	CONSRH	Uses concentration dependent lifetimes. Recommended for Si.
Auger	AUGER	Direct transition of three carriers. Important at high current densities.
Optical	OPTR	Band-band recombination. For direct materials only.
Surface	S.N S.P	Recombination at semiconductor to insulator interfaces. Set on the INTERFACE statement.

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Table 2-4. Impact Ionization		
Model	**Syntax**	**Notes**
Silberrherr's Model	IMPACT SELB	Recommended for most cases. Includes temperture dependent parameters.
Grant's Model	IMPACT	Similiar to Selberrherr's model but with different coefficients.
Crowell-Sze	IMPACT CROWELL	Uses dependence on carrier scattering length.
Toyabe Model		Non-local model used with Energy Balance. Any IMPACT syntax is accepted.
Concannon	N.CONCAN P.CONCAN	Non-local model developed in Flash EEPROM technologies.

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Table 2-5. Tunneling Models and Carrier Injection Models		
Model	**Syntax**	**Notes**
Fowler-Nordheim (electrons)	FNORD	Self-consistent calculation of tunneling through insulators. Used in EEPROMs.
Fowler-Nordheim (holes)	FNHOLES	As FNORD for holes. Not usually required.
Band-to-Band (standard)	BBT.STD	For direct transitions. Required with very high fields.
Klaassen Band-to-Band	BBT.KL	Includes direct and indirect transitions.
Hot Electron Injection	HEI	Models energetic carriers tunneling throug insulators. Used for gate current and Flash EEPROM programming.
Hot Hole Injection	HHI	As HEI for holes. For ULSI PMOS devices.
Concannon Gate Current Model	N.CONCAN P.CONCAN	Non-local gate model consistent with Concannon substrate current model.

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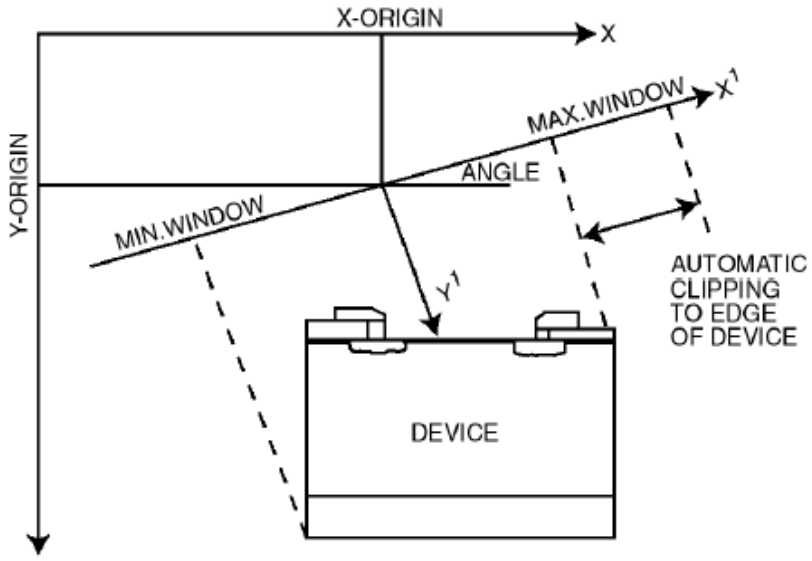
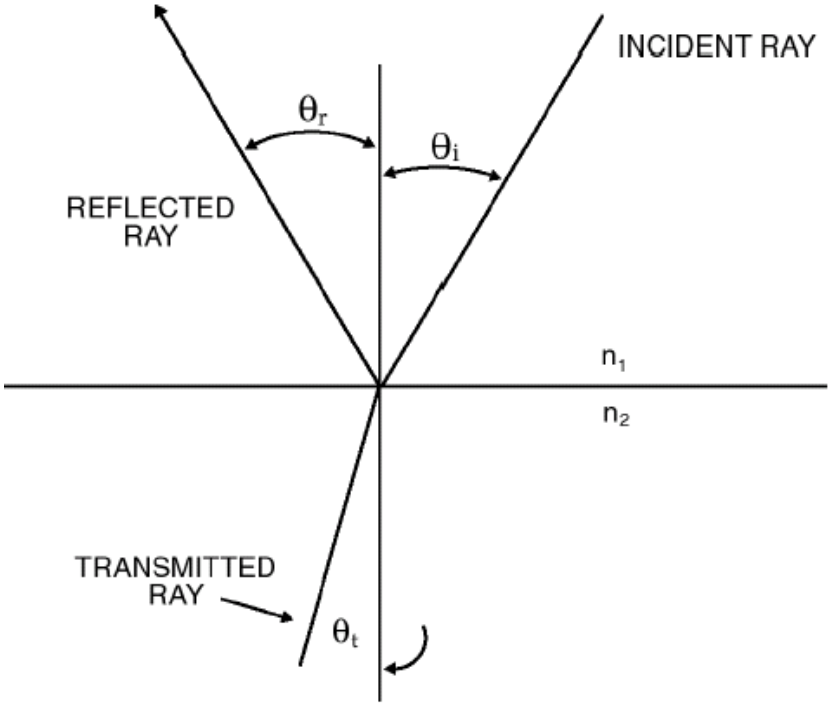
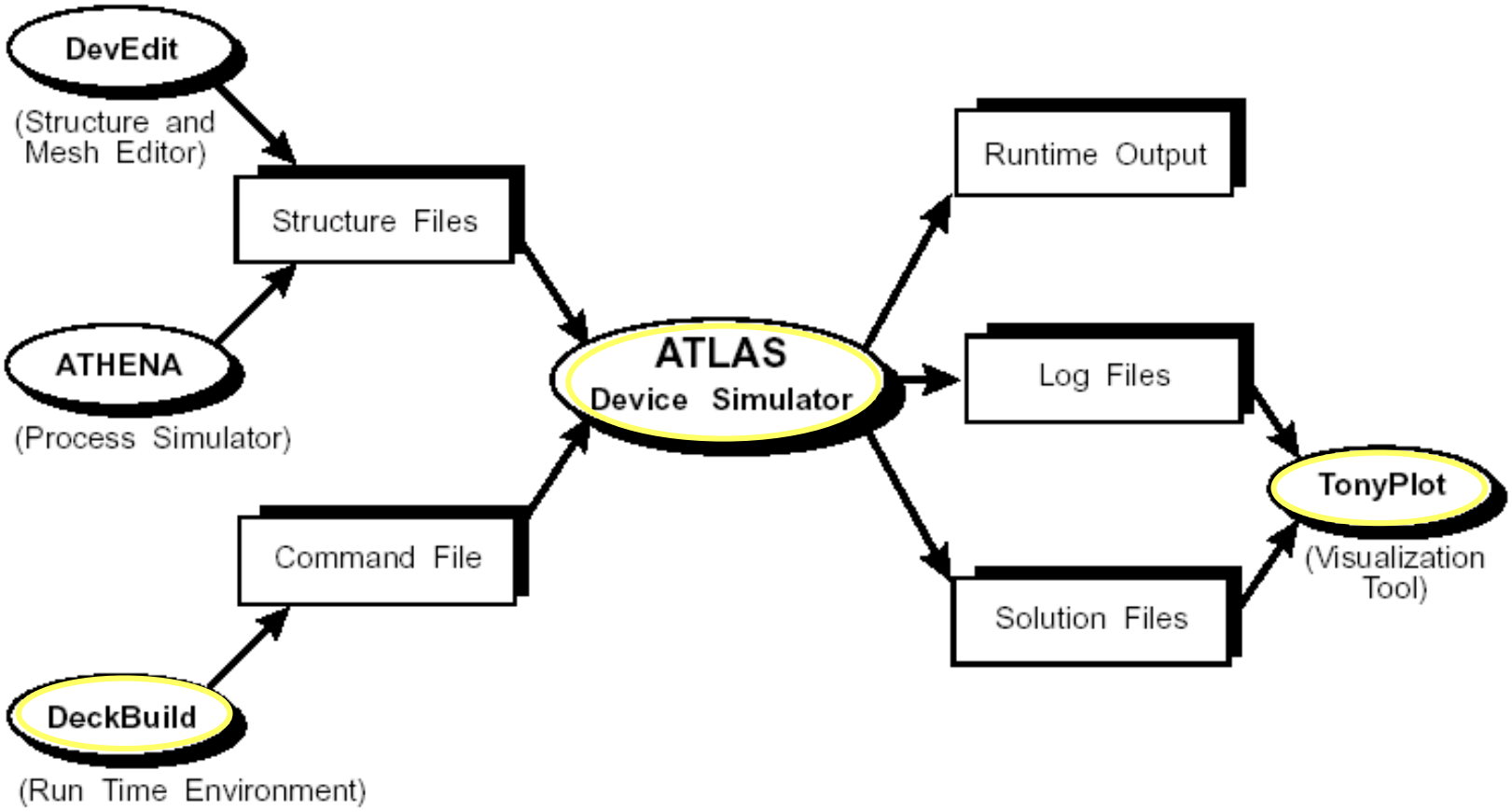


Figure 8-1: Optical Beam Geometry



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struktura vstupního souboru

Group	*Statements*
1. Structure Specification	MESH REGION ELECTRODE DOPING
2. Material Models Specification	MATERIAL MODELS CONTACT INTERFACE
3. Numerical Method Selection	METHOD
4. Solution Specification	LOG SOLVE LOAD SAVE
5. Results Analysis	EXTRACT TONYPLOT

```
go atlas
mesh
x.mesh loc=0.00 spac=0.5
x.mesh loc=3.00 spac=0.2
x.mesh loc=5.00 spac=0.25
x.mesh loc=7.00 spac=0.25
x.mesh loc=9.00 spac=0.2
x.mesh loc=12.00 spac=0.5
y.mesh loc=0.00 spac=0.1
y.mesh loc=1.00 spac=0.1
y.mesh loc=2.00 spac=0.2
y.mesh loc=5.00 spac=0.4
region  num=1  silicon
electr  name=anode  x.min=5  length=2
electr  name=cathode  bot
#....   N-epi doping
doping  n.type conc=5.e16 uniform
#....   Guardring doping
doping   p.type conc=1e19 x.min=0  x.max=3  junc=1 rat=0.6 gauss
doping   p.type conc=1e19 x.min=9 x.max=12 junc=1 rat=0.6 gauss
#....   N+ doping
doping  n.type conc=1e20 x.min=0 x.max=12 y.top=2 y.bottom=5 uniform
save outf=diodeex01_0.str
model    conmob  fldmob  srh  auger  bgn
contact    name=anode workf=4.97
solve      init
method newton
log   outfile=diodeex01.log
solve      vanode=0.05  vstep=0.05  vfinal=1  name=anode
tonyplot diodeex01.log
quit
```